Practical 5: Installing KoboToolbox & Designing a Survey

Day 3 — Duration: 3 Hours

SDI Training for ICT Professionals — Climate Risk Database

What you will accomplish: By the end of this practical, you will have KoboToolbox running on your laptop, a flood assessment survey form designed and deployed, and test data submitted with GPS coordinates. KoboToolbox is a free tool for mobile data collection widely used in humanitarian and development work.

Before you start: Docker Desktop must be running. You need at least 4 GB of free RAM and 10 GB of free disk space for KoboToolbox (in addition to what CRD is using). If your laptop has only 8 GB of RAM, the instructor may ask you to stop CRD first: `docker compose down` from the `my-crd` folder.

Part 1: Installing KoboToolbox (2 Hours)

Download the KoboToolbox Installer

Open your terminal:

- **Windows:** Open **PowerShell**
- **Mac:** Open **Terminal**
- **Ubuntu:** Press `Ctrl+Alt+T`

Step 1: Navigate to your home folder:

```
cd ~
```

What this does: Makes sure you start from your home folder.

You should see: No output.

Step 2: Download the KoboToolbox installer:

```
git clone https://github.com/kobotoolbox/kobo-install.git
```

What this does: Downloads the KoboToolbox installation scripts from GitHub. These scripts automate the complex process of setting up KoboToolbox with Docker.

You should see:

```
Cloning into 'kobo-install'...
remote: Enumerating objects: ...
Resolving deltas: 100% (...), done.
```

Step 3: Enter the installer folder:

```
cd kobo-install
```

What this does: Moves into the folder containing the installation scripts.

You should see: No output.

Run the Installation Wizard

Step 4: Start the installation wizard:

```
python3 run.py
```

What this does: Launches an interactive setup wizard that asks you questions about how you want to configure KoboToolbox. You answer each question by typing your choice and pressing Enter.

You should see: The wizard begins asking questions. Here are the answers to give for each question:

Wizard answers — type exactly what is shown:

Q1: "What kind of installation do you want?"
Type: `1` (for "On your local machine") → Press Enter

Q2: "Do you want to use HTTPS?"
Type: `no` → Press Enter

Q3: "What is the hostname?"
Type: `localhost` → Press Enter
(or accept the default if `localhost` is suggested)

Q4: "What port for KPI (frontend)?"
Accept the default (usually `8080`) → Press Enter

Q5: "What port for KoBoCAT (backend)?"
Accept the default (usually `8000`) → Press Enter

Q6: "What port for Enketo Express?"
Accept the default (usually `8005`) → Press Enter

Q7: "Create a superuser?"
Type: `yes` → Press Enter

Q8: "Superuser username:"
Type: `admin` → Press Enter

Q9: "Superuser password:"
Type: `admin123` → Press Enter
(This is only for training — use a strong password on real servers)

Q10: "Run the setup now?"
Type: `yes` → Press Enter

Note: The exact questions may vary slightly depending on the version of kobo-install. The instructor will guide you through any differences. The key points are: local installation, no HTTPS, localhost, and create a superuser account.

Step 5: Wait for the setup to complete. The wizard will download Docker images and configure the services:

```
Pulling images ...
Creating containers ...
Setting up database ...
```

What this does: Docker downloads all the KoboToolbox components (web server, database, form engine, etc.) and sets them up.

You should see: Many lines of download and setup progress. This takes **10–20 minutes** depending on your internet speed.

Start KoboToolbox

Step 6: If KoboToolbox did not start automatically, start it manually:

```
python3 run.py --start
```

What this does: Starts all KoboToolbox Docker containers.

You should see:

```
Starting KoBoToolbox...
[+] Running 8/8
 - Container kobo-postgres Started
 - Container kobo-mongo Started
 - Container kobo-redis Started
 - Container kobo-kpi Started
 - Container kobo-kobocat Started
 - Container kobo-enketo Started
 - Container kobo-nginx Started
 ...
```

Step 7: Wait for all services to finish starting (about 3–5 minutes after the containers start). You can check readiness:

```
docker ps --format "table {{.Names}}\t{{.Status}}"
```

What this does: Shows a table of all running Docker containers and their status.

You should see: All KoboToolbox containers listed with status "Up" and a time indicator. Wait until all show "Up" for at least 1 minute.

Step 8: Open your browser and go to:

```
http://localhost:8080
```

What this does: Opens the KoboToolbox web interface running on your laptop. Port 8080 is the default for KoboToolbox's frontend.

You should see: The KoboToolbox login page with the KoboToolbox logo and a "Log In" form.

If you see a blank page or error: Wait 2–3 more minutes and refresh. KoboToolbox takes time to fully initialise. If it still does not work after 5 minutes, check the logs:

```
docker logs kobo-kpi --tail 30
```

Step 9: Log in with the superuser account you created during the wizard:

- **Username:** `admin`
- **Password:** `admin123` (or whatever you chose)

You should see: The KoboToolbox dashboard — a clean interface with a "New" button and an empty project list. This means KoboToolbox is fully working.

Congratulations! KoboToolbox is installed and running. The hard part (installation) is done. Now the fun part — designing a survey form.

Part 2: Designing a Flood Assessment Survey (1 Hour)

You will now design a survey form for collecting flood damage data in the field. This form uses the **visual form builder** — no coding or XLSForm knowledge needed.

Create a New Form

Step 10: On the KoboToolbox dashboard, click the blue **"New"** button.

Step 11: Select **"Build from scratch"** (or **"Build form"**).

You should see: The form builder opens. It has a blank area in the centre where you will add questions, and a toolbar or sidebar with question types.

Step 12: At the top, you will see a field for the form name. Click it and type:

```
Flood Damage Assessment - Tanga 2026
```

Add Survey Questions

You will now add 7 questions to the form. For each question, click the **"+"** button (or drag a question type from the sidebar) and configure it.

Step 13: Question 1 — GPS Location:

- Click **"+"** to add a new question
- Select question type: **"Point"** (under Location/GPS)
- Label: `GPS Location of Assessment`
- This will automatically capture GPS coordinates when the surveyor fills in the form

You should see: A GPS/Point question block appears in the form.

Step 14: Question 2 — Flood Depth:

- Click **"+"** to add a new question
- Select question type: **"Number"** (or "Integer")
- Label: `Flood Water Depth (centimetres)`
- In the question settings, you can set:
  - Minimum value: `0`
  - Maximum value: `500`

You should see: A number input question appears below the GPS question.

Step 15: Question 3 — Date of Flooding:

- Click **"+"**
- Select question type: **"Date"**
- Label: `Date of Flooding`

You should see: A date picker question appears in the form.

Step 16: Question 4 — Ward Name:

- Click "+"
- Select question type: **"Select One"** (dropdown or radio buttons)
- Label: `Ward Name`
- Add these choices (one per line):
  - `Chumbageni`
  - `Makorora`
  - `Ngamiani`
  - `Mzingani`
  - `Tanganyika`
  - `Other`

You should see: A multiple-choice question with 6 options.

Step 17: Question 5 — Photo of Damage:

- Click "+"
- Select question type: **"Photo"** (or "Image")
- Label: `Photo of Flood Damage`

You should see: A photo capture question. On a mobile phone, this will open the camera.

Step 18: Question 6 — Households Affected:

- Click "+"
- Select question type: **"Number"** (or "Integer")
- Label: `Number of Households Affected`
- Set minimum value: `0`

You should see: A number input question.

Step 19: Question 7 — Type of Damage:

- Click "+"
- Select question type: **"Select Multiple"** (checkboxes — multiple answers allowed)
- Label: `Type of Damage Observed`
- Add these choices:
  - `Structural damage to buildings`
  - `Road damage or blockage`
  - `Crop or farmland damage`
  - `Contaminated water supply`
  - `Displacement of residents`
  - `Loss of personal property`

You should see: A checklist question where surveyors can select multiple damage types.

Step 20: Review your form. You should see all 7 questions listed in order. Check that the labels are correct and question types are right.

> **Tip:** You can reorder questions by dragging them up or down. You can also click the settings icon on any question to mark it as "Required" (the surveyor must answer it before submitting).

Deploy the Form

Step 21: Click the **"Save"** button (usually top-right) to save your form.

You should see: A confirmation that the form is saved.

Step 22: Click the **"Deploy"** button (this may also be labelled "Deploy this version" or show as a rocket icon).

You should see: A confirmation dialog asking if you want to deploy. Click **"Deploy"** or **"OK"**.

Step 23: After deployment, you should see the form status change to **"Deployed"**. The form is now live and ready to accept submissions.

You should see: The form page now shows tabs for "Form", "Data", "Settings", and a link to the web form.

Submit Test Entries

Step 24: On the form page, look for an **"Open"** button or click on the web form link (it may say "Preview" or show an eye icon for the Enketo web form).

Step 25: Alternatively, look for the **"Collect data"** tab or button, and select **"Open in web browser"** (Enketo).

You should see: The survey form opens in a new browser tab, ready to be filled in. It looks like a clean questionnaire.

Step 26: Fill in **Test Entry 1** with these values:

- **GPS Location:** Since you are at a computer (not in the field), click the map and place a point near Tanga city centre. Use approximate coordinates: **Latitude: -5.069**, **Longitude: 39.096**
- **Flood Water Depth:** `45`
- **Date of Flooding:** Select today's date or any recent date
- **Ward Name:** `Chumbageni`
- **Photo:** Skip this for now (or take a photo of anything nearby)
- **Households Affected:** `12`
- **Damage Type:** Check `Structural damage to buildings` and `Road damage or blockage`

Step 27: Click **"Submit"** at the bottom of the form.

You should see: A success message: "Your data has been submitted successfully."

Step 28: Submit **Test Entry 2** with different values:

- **GPS:** Place the point slightly north: **Latitude: -5.060**, **Longitude: 39.100**
- **Flood Depth:** `80`

- **Date:** A different date
- **Ward:** `Ngamiani`
- **Households:** `25`
- **Damage:** Check `Contaminated water supply` and `Displacement of residents`

Click **"Submit"**.

Step 29: Submit **Test Entry 3** with your own values:

- Place the GPS point somewhere else near Tanga
- Choose different answers for each question
- Click **"Submit"**

View the Collected Data

Step 30: Go back to the KoboToolbox dashboard. Click on your form name: **"Flood Damage Assessment - Tanga 2026"**.

Step 31: Click on the **"Data"** tab.

You should see: A data table showing all 3 submissions. Each row is one submission, and columns show the answers to each question.

Step 32: Explore the data views:

- **Table view:** Shows data in a spreadsheet-like table
- **Map view:** Click the map icon to see your 3 GPS points on a map
- **Reports:** Click to see summary charts and graphs

You should see: Three points on the map near Tanga. Clicking a point shows the submission details.

In real field work: Surveyors would use the KoboCollect mobile app (Android) or Enketo web forms (any device) to submit data while in the field. GPS coordinates are captured automatically from the phone's GPS. Photos are taken with the phone camera. Data syncs to the server when internet is available.

Checkpoint

Before proceeding to Practical 6, confirm:

☐ **KoboToolbox running:** Login page loads at `http://localhost:8080`

☐ **Admin login works:** You can sign in with your admin account

☐ **Form created:** "Flood Damage Assessment - Tanga 2026" exists

☐ **Form deployed:** The form status shows "Deployed"

☐ **Test data submitted:** 3 entries with GPS coordinates are visible in the Data tab

☐ **Map view works:** 3 points appear near Tanga on the data map view

Excellent work! You now have a complete field data collection system. In the next practical,

you will export this data and upload it to CRD — connecting field collection to your spatial data platform, all without writing a single line of code.

End of Practical 5 — Proceed to Practical 6: From KoboToolbox to CRD